Rating the Pitch, Not the Product: User Evaluations of LLMs Reflect Expectations More Than Performance

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Abstract

Imagine two users interact with the same LLM. One is told it is the cutting-edge flagship; the other, an older, weaker model. They walk away with markedly different ratings of its usefulness and intelligence, yet they used the *same model*. In a controlled study, 162 participants each used one of six LLMs from two families across three collaborative tasks, after first viewing a landing page that matched, overstated, or understated their model's true capability. This pre-interaction *framing* persisted through use. Oversold users rated the model more favorably than Undersold users before interaction; afterward, both groups had updated toward reality, but the impression gap did not fully close. Framing also changed how users worked with the model (with Oversold users prompting more directively, Undersold users more collaboratively), but the quality of what they produced together depended only on the model's true capability, not on what users were told. Participants' change in model impressions after use, measured across two impression measures, was *not* predicted by task performance ($\beta = -0.01$ and 0.11, both n.s.), but by whether the model met users' expectations ($\beta = 0.47$ and 0.50, both $p < .001$) and how confident they felt working with it ($\beta = 0.47$ and 0.36, both $p < .001$). After interaction, users are still rating the *pitch*, not the *product*: user-elicited LLM evaluations, including the preference data driving public leaderboards, measure expectation management at least as much as the model itself.

1 Introduction

Most users meet an LLM before they *actually* meet it. They might have seen a benchmark score, a launch post, a leaderboard ranking (Chiang et al., 2024), or heard a colleague's recommendation. These pre-interaction signals may have lasting consequences for how users evaluate, adopt, and trust the systems they go on to use. Do initial expectations fade as users accumulate experience, or do they persist? Do they only affect impressions, or also how users interact with the model and what they produce together? And when impressions do change, what drives that change: the quality of the output, or something else about the experience?

Pre-interaction expectations are well-known drivers of technology evaluation, adoption, and abandonment (Bhattacherjee, 2001; Budhathoki et al., 2024), and post-interaction impressions shape satisfaction often independently of objective performance (Blut et al., 2021; Avcılar and Yenilmez, 2026). As LLMs become embedded in workflows where mismatched expectations can drive misuse or misplaced trust (Noy and Zhang, 2023), how expectations evolve into impressions through use is consequential.

Work in human-AI interaction has begun to ask these questions. The *framing* of an AI system (the description users encounter before use) shifts judgments of trustworthiness (Pataranutaporn et al., 2023), misjudged capability changes what users produce (Bangerl et al., 2025), and passive rather than collaborative use changes their sense of ownership over the output (Lee et al., 2026). No prior work, however, has followed users across iterative LLM use to ask whether these effects persist or generalize to behavior and output, and what sustains them. For LLMs, where users engage in multi-turn collaboration and capability is signaled through public benchmarks (Kocielnik et al., 2019), marketing, and word of mouth, these questions matter for individual users and the preference-based evaluation pipelines that aggregate their judgments.

We ran a controlled between-subjects study with 162 participants, each randomly assigned a **source** model (the model actually serving the participant) and a **shown** model (the model described to them), which determines framing condition (*Oversold*, *Matched*, and *Undersold*), from two model families (GPT & Claude). Each first saw a landing

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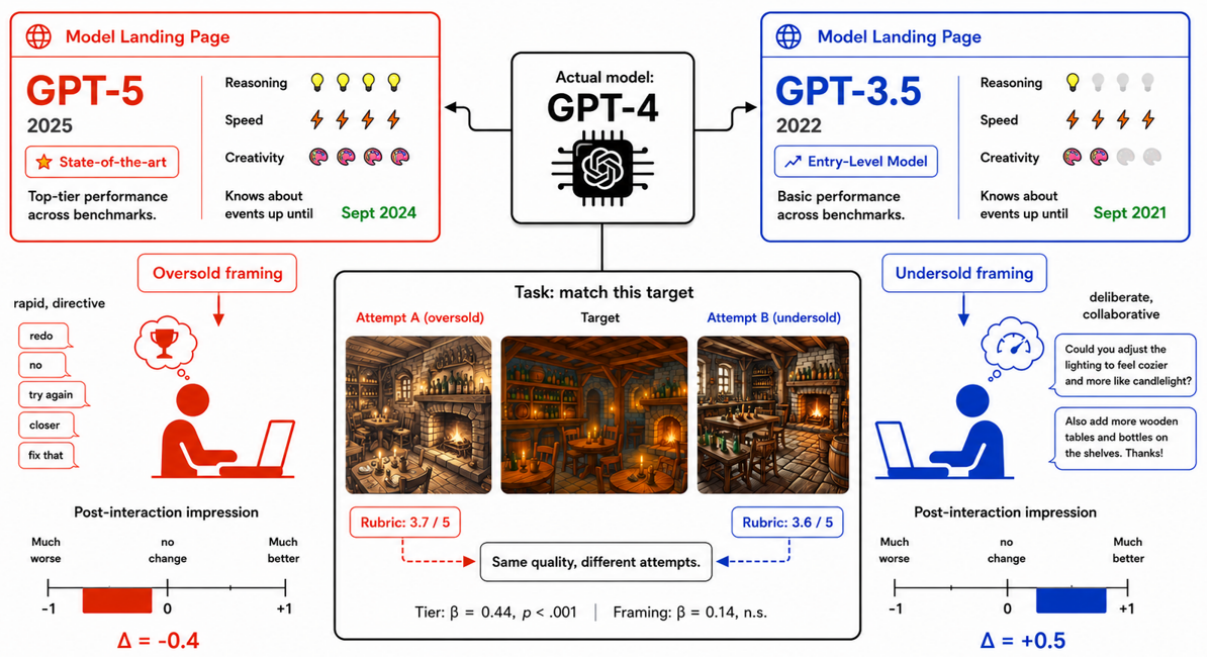


Figure 1: **Same model, different framings, different user experience.** Illustrated with GPT-4 framed as either state-of-the-art (Oversold) or entry-level (Undersold) using a real task and outputs from the study; the full study spans six models across GPT and Claude families. Framing changed user interaction behavior and post-interaction impressions, but not task performance.

page introducing a model from their assigned family, which did not always match the model actually serving them in the backend (Appendix Figure 10): Matched saw their true model, Oversold saw a higher-tier model (e.g., they were *using* GPT-3.5 but were *shown* GPT-5), and Undersold saw a lower-tier model (e.g., they were *using* GPT-5 but were *shown* GPT-3.5). All completed three collaborative tasks (image generation, outreach message writing, and acronym building). We measured impressions before and after the session, task-specific confidence around each task, interaction behavior from conversation logs, and output quality. Figure 1 summarizes the design and central results.

We establish three findings:

- **The impression gap set by framing does not close with use.** Oversold participants entered with higher impressions and Undersold with lower; after three tasks of hands-on work, both groups updated towards reality, but the gap did not fully close, though expectation-disconfirmation accounts predict that experience should wash the initial framing out (§3.1).
- **Framing does not change the quality of what users and the model produce together, only how they work to produce it.** Under different framings, the same model produced different interaction behavior: Oversold users prompted more directively, Undersold users more collaboratively, with the divergence largest on the acronym task, however task performance did not vary with framing (§3.2, §3.3).
- **Impression change is driven by experience, not output.** What predicts how users' impressions of an LLM shift after using it is not their task performance ($\beta = -0.01$ and 0.11, both n.s.), but whether the model met their expectations and how their task-specific confidence (i.e., their *self-efficacy*) changed ($\beta = 0.36$ to 0.50, all $p < .001$) (§3.4).

The implications are broad. Users' impressions of a model are partly set before any prompt is sent. For organizations deploying AI tools, *how* a tool is introduced may matter as much as the tool itself. For developers and benchmark designers, expectation-setting signals (benchmarks, marketing, interfaces) appear to drive user impressions more than capability differences users detect during use, particularly for open-ended tasks (e.g., those involving creativity, expression, and opinion). For evaluation research, the concern is one of measurement validity: preference data elicited from users may measure expectation management at least as much as model capability.

Source tiers (rows) vs. Shown tiers (columns)	Bottom	Middle	Top
Bottom	Matched	Oversold	Oversold
Middle	Undersold	Matched	Oversold
Top	Undersold	Undersold	Matched

Table 1: Framing condition, determined by the relation between the model serving the participant (source tier) and the model described on the landing page (shown tier). Each combination was run in both model families.

Tier	Source model	Shown name
GPT family		
Bottom	`gpt-3.5-turbo-0125`	GPT-3.5
Middle	`gpt-4-0125-preview`	GPT-4
Top	`gpt-5-2025-08-07`	GPT-5
Claude family		
Bottom	`claude-3-haiku-20240307`	Claude 3
Middle	`claude-3-5-haiku-20241022`	Claude 3.5
Top	`claude-sonnet-4-20250514`	Claude 4

Table 2: Source models used in the study. Each source model was paired with each shown name in its family across participants, defining the framing manipulation.

2 Methodology

To test whether pre-interaction framing of an LLM's capability persists into use, and what drives impressions if it does, we crossed three source tiers with three shown tiers, across two model families: GPT (Ouyang et al., 2022; OpenAI, 2023, 2025) and Claude (Anthropic, 2024a,b, 2025). Framing (*Oversold*, *Matched*, *Undersold*) is the relation between the two. 162 participants completed three collaborative tasks of varying objectivity; we measured impressions of the model before and after the session, self-efficacy before and after each task, expectations met after each task, and the quality of the final output, while tracking behavioral metrics.

2.1 Capability Framing

Each participant was introduced to their assigned model through a landing page before any interaction began (Appendix Figure 10). The page displayed the model's name, release year, knowledge cutoff, and three capability scores (reasoning, speed, and creativity) on a 1–4 scale, alongside a comparison against other models in its family. The reasoning and creativity scores were derived from the model's percentile rank on LMArena's Hard Prompts and Creative Writing leaderboards, respectively,[1] at the time of the study; the speed score was derived from the latency metric from each provider's model overview page.[2] The capability tier shown did not always correspond to the model actually served by the backend.

Each family had three models at capability tiers ordered by public benchmark rankings at the time of data collection (Chiang et al., 2024). Table 2 lists the six source models and the shown names within each family, where every source model was paired with every shown tier, yielding nine source-by-shown combinations per family and 18 across the full design. The framing condition follows from this pairing rather than being assigned separately: a participant is Oversold when the shown tier sits above the source tier, Undersold when it sits below, and Matched when the two agree (Table 1). Nine participants were assigned to each combination, for N=162, balanced at 54 per framing condition, 81 per family, and 27 per source model. These numbers were determined by a pilot of 18 participants (one per source-by-shown cell) and a Monte Carlo power analysis targeting 80% power at $\alpha = 0.05$.

Testing whether output quality tracks real capability while impressions track the description requires true and described capability to vary independently. Three tiers per family supply that variation, and repeating the manipulation in a second family checks that the results are not specific to one lineage.

The pairing also creates a distance between source and shown tier, so we use framing extent (source tier minus shown tier, -2 to $+2$) to test for dose-response effects.

2.2 Tasks

Participants completed three tasks in the same order: image generation, outreach message writing, and acronym building.

Image generation. Participants worked with the model to generate an image resembling the target image as closely as possible (Appendix Figure 14). DALL-E 3 rendered every image, with the source model refining the participant's request into the prompt, so differences between source tiers enter this task only through prompt quality.

Outreach message writing. The task was to draft a persuasive outreach message for a sales role application (Appendix Table 8). The message had to fulfill several requirements: (a) reference the job description, (b) be signed "J. Doe", (c) be addressed to "Dr. Rogers", and (d) read as if written by a human rather than an AI tool.

[1] https://arena.ai/leaderboard/

[2] Anthropic: Claude models overview; OpenAI: OpenAI models overview

Acronym building. The goal was to generate humorous acronyms from a set of letters (e.g., '**PUGT**' → '**P**rolific **U**sers **G**et **T**ired'). They worked with three letter sets: (a) B L M P F, (b) C R T W, and (c) Y O L D. With no objectively correct answer or specific requirements to fulfill, quality depended entirely on the participant's creative direction in collaboration with the model.

All tasks required sustained, iterative collaboration with the model rather than single-turn prompting. The tasks were chosen to reflect varying levels of objectivity: the first has a fixed external target, the second is anchored to a written brief, and the third is fully open-ended and subjective. This spread allows us to observe whether framing effects vary with how much the user's own creative input shapes what counts as a good output.

Participants were told each task required at least five minutes of interaction and the highest-scoring submission per task would earn a $10 bonus.

2.3 Measures

Impressions. We used the Unified Theory of Acceptance and Use of Technology (UTAUT) Performance Expectancy and Effort Expectancy subscales (Venkatesh et al., 2003), which capture usefulness and ease of use respectively (Appendix Table 5), and the Godspeed Perceived Intelligence subscale (Bartneck et al., 2009), which captures perceived intelligence (Appendix Table 7). Both were administered before the study (future tense) and after (past tense).

Self-efficacy. Self-efficacy refers to a person's belief in their ability to perform a task (Chen et al., 2001). We adapted the New General Self-Efficacy scale into eight task-specific items, completed before and after each task, to capture how participants' self-efficacy in working with the model changed across the study (Appendix Table 6).

Expectations met. After each task, participants rated if the model fell short, met, or exceeded their expectations for that task on a three-point scale.

Behavior. From each participant's task conversation logs, we extracted 17 behavioral metrics, covering message counts and lengths, keystroke and backspace counts, edit events, session and idle time, and time between messages (full list in Appendix Table 15). Each user message was classified as *collaborative* or *directive* by `claude-haiku-4-5`, using a prompt grounded in Searle (1976)'s illocutionary taxonomy and Roschelle and Teasley (1995)'s work on collaborative engagement (prompt and method in Appendix A.5). Human annotators validated the classifier on a stratified sample of messages, and its agreement with them was fair against their own moderate agreement (Appendix A.8).

Output quality. Participants selected and submitted one final product per task, which was scored by GPT-5 against a task-specific rubric on a 1–5 scale across four dimensions, averaged over three scoring runs (Rubric in Appendix Table 9; full prompt and method in Appendix A.6). Human raters validated the judge on a subset of submissions, where its correlation to humans across tasks was similar to humans' correlation to each other (Appendix A.7).

Examples of high- and low-scoring submissions for image generation are shown in Figures 12 and 13, for outreach messages in Table 10, and for acronyms in Table 11, of Appendix A.4.

2.4 Participants and Procedure

Recruitment. We recruited 162 US-based workers through Prolific, all of whom were fluent in English, had a 98–100% approval rate, and had completed more than 100 prior studies. Alongside the study measures we collected each participant's chatbot use frequency and self-reported comfort with chatbots. Assignment to the 18 source-by-shown cells was managed by an SQL database that atomically reserved an open cell for each incoming participant, preventing race conditions and maintaining the balanced design across the full data collection period. All study procedures were approved by the authors' institutional research ethics board prior to data collection, and all participants provided informed consent before beginning the study.

Procedure. The study was administered across Prolific, Tally, and a custom web-based chat interface powered by the source LLM (Appendix Figure 11). After consent and pre-study UTAUT and Godspeed surveys, participants viewed the landing page (manipulation) and had to pass a 3-item manipulation check. They then completed the three tasks in sequence, each bracketed by the self-efficacy scale, and rated whether the model met their expectations after each. Post-study UTAUT and Godspeed surveys closed the session.

Data integrity. To detect automated or low-effort participation, we implemented layered checks.

Measure	Condition	Mean	SD
Pre-UTAUT	Oversold	4.21	0.65
	Matched	4.05	0.73
	Undersold	3.96	0.54
Pre-Godspeed	Oversold	5.46	1.05
	Matched	5.33	1.04
	Undersold	4.79	0.97

Table 3: Pre-interaction UTAUT (1–5; higher = more useful and easier to use) and Godspeed Perceived Intelligence (1–7; higher = more intelligent and competent) by framing condition. N=54 per condition.

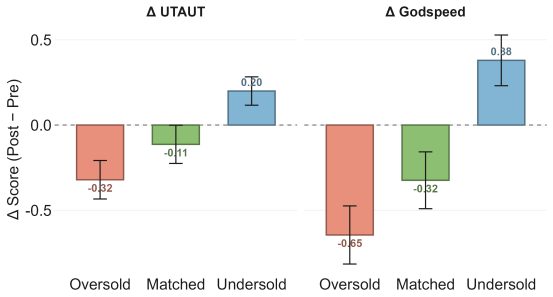


Figure 2: Mean impression change (Δ = post − pre) on UTAUT (left) and Godspeed (right) by framing condition. Error bars are ± 1 bootstrap SE (B=5000).

Each task page contained a honeypot button invisible to human users but detectable by browser automation agents, hidden DOM elements with task-specific seeded keywords later searched in conversation logs, and browser-environment checks flagging known indicators of automation. Three attention check items were embedded across the surveys (Appendix A.1). All submissions were then reviewed manually, blind to framing condition and before any outcome analysis; ten participants were removed and replaced through new recruitment, preserving the balanced N=162 (criteria and counts in Appendix A.2).

2.5 Analysis

Codings and outcomes. Analyses were conducted in R. Framing was coded $\{-1, 0, +1\}$ (Under/Matched/Over) and model tier $\{-1, 0, +1\}$ (Bottom/Middle/Top); framing extent was used as an integer covariate (-2 to $+2$). Impression scores are mean item ratings for the UTAUT and Godspeed respectively; impression change scores are within-participant differences, each person's own post composite minus their own pre composite (i.e., $UTAUT_{post} - UTAUT_{pre}$ and $Godspeed_{post} - Godspeed_{pre}$). Self-efficacy change is computed per task ($SE_{post} - SE_{pre}$) and averaged across the three tasks. Overall performance is the equal-weighted mean of the three task-level judge scores.

Statistical tests. We used independent-samples t-tests with Cohen's d for pre-study impression contrasts (§3.1). Bar-chart figures show group means with 95% CIs. Ordinary Least Squares (OLS) regression was used for the framing-extent dose-response model (§3.1, Figure 3, jointly with source tier) and the performance regression on framing and model tier (§3.3). For §3.4, we ran six bivariate OLS regressions of each impression-change outcome on three predictors: task performance, self-efficacy change, and expectations met.

Multiple comparisons. All p-values are reported uncorrected. In place of an α-adjustment, each effect of theoretical interest is tested on two impression measures drawn from different validated instruments, and in §3.4 against two independent experiential predictors; all central effects replicate across both (more information in §5).

3 Results

3.1 Framing sets an impression gap that use does not close

Before interaction, framing produced a clear ranking of impressions. Oversold participants entered with the highest ratings (UTAUT M=4.21, Godspeed M=5.46) and Undersold the lowest (M=3.96 and M=4.79), with Matched in between (Table 3). The contrast between Oversold and Undersold was significant on both pre-UTAUT (p=0.029, d=0.43) and pre-Godspeed (p=0.0009, d=0.66) (pairwise contrasts in Appendix Table 16). The effect was larger on Godspeed than on UTAUT, indicating that perceived intelligence is more sensitive to framing than expected performance or effort.

Expectation-disconfirmation accounts predict that users revise their expectations toward reality as they accumulate experience (Oliver, 1980; Bhattacherjee, 2001), so three tasks of hands-on collaboration should have closed the gap that framing opened. It narrowed but did not close: ratings shifted toward the models' actual performance yet remained differentiated across conditions (Figure 2). Undersold participants updated upward, Oversold updated downward, and Matched drifted slightly downward, suggesting that even an accurately described model can mildly underwhelm. The pattern held for both scales (Appendix Figure 18).

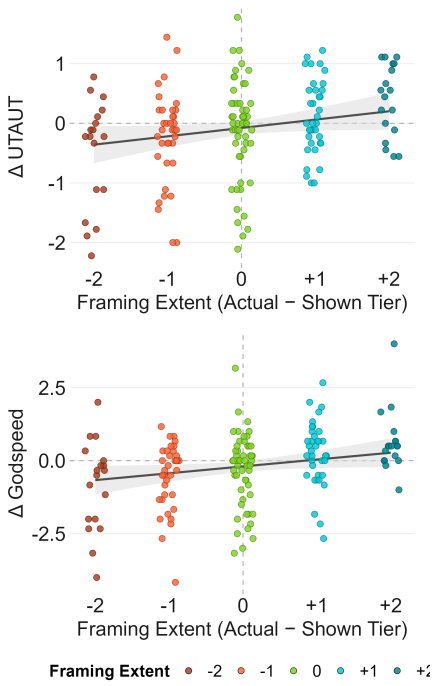


Figure 3: OLS regression of impression change on framing extent (source tier − shown tier), with source tier held at its mean. Points are jittered participants; band is the 95% CI on the regression line.

Impression change followed a dose-response relationship with framing extent: controlling for source tier, both Δ_{UTAUT} and Δ_{Godspeed} rose monotonically (Figure 3). The more Undersold the framing, the larger the upward update; the more Oversold, the larger the downward update.

The size of these framing effects did not vary with participants' prior familiarity: neither chatbot use frequency nor self-reported comfort with AI moderated framing's effect on pre-interaction impressions or on impression change (Appendix Table 17).

3.2 Framing changes interaction behavior

Framing also changed how participants worked with the model. On the acronym task, Oversold participants sent the most messages, with the shortest gaps and shortest prompts, consistent with re-prompting in search of an output the model was failing to deliver; Undersold participants sent fewer, longer messages with more deliberation between them, consistent with active co-construction; Matched fell between the two on every metric (Figure 4). Read in task order, the gradient is weak and inconsistent on image generation (Appendix Figure 23), attenuated on outreach message writing (Appendix Figure 24), and steepest on acronym building: the divergence grows across the session, and only the acronym task left user input as the primary determinant of a good output.

The same split appears in what participants wrote and in what they submitted. On the acronym task, Undersold participants sent markedly more collaborative messages than Oversold, with Matched in between, while the proportion varied little with framing on the other two tasks (Figure 5). Oversold participants' acronym submissions were also roughly twice as close to the model's own wording as Undersold participants', measured as normalized Levenshtein similarity to the most similar assistant message in their log (0.32 against 0.17, p=.001; Appendix Table 18); on outreach the measure was near ceiling across conditions, since the model's draft already met the brief's requirements. Representative chat logs are in Appendix Figures 16 and 17.

Participants' own reports after each task follow the same order: the framing gap in self-efficacy change and in expectations met is small on image generation and outreach, then widens on acronym building, where the difference in expectations met reaches significance (Appendix Table 19).

3.3 Output quality corresponds to model capability, not framing

Framing did not affect the quality of what users and the model produced together. Performance increased linearly with model tier and was flat across framing conditions within each tier (Figure 6). Oversold participants did not produce better outputs because they expected a better model and Undersold participants did not produce worse ones because they expected a weaker one.

A regression of z-scored performance on framing and model tier confirms this. Model tier predicted performance (β=0.44, p=0.0006) and framing did not (β=0.14, p=0.276). A two-way ANOVA treating both as categorical factors gives the same separation without assuming the tier effect is linear (tier F=6.12, p=0.003, η_p^2=0.072; framing F=0.68, p=0.51; Appendix Table 20). The same pipeline therefore registers real capability differences while registering none across framing. The model tier gradient appeared on two of the three tasks (Figure 7), reaching significance on outreach message writing and acronym building; on image generation, where every image was rendered by the same backend model, scores clustered tightly and neither factor separated the groups.

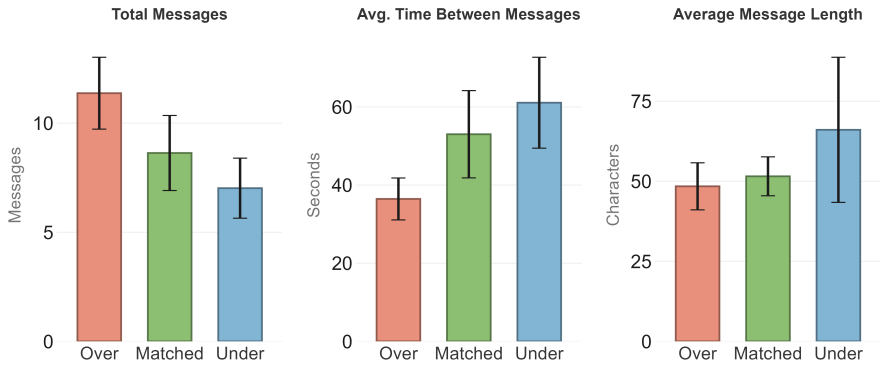


Figure 4: Three engagement metrics on the acronym task by framing condition (mean ±95% CI). From left to right: total messages sent, mean time between messages, and mean message length.

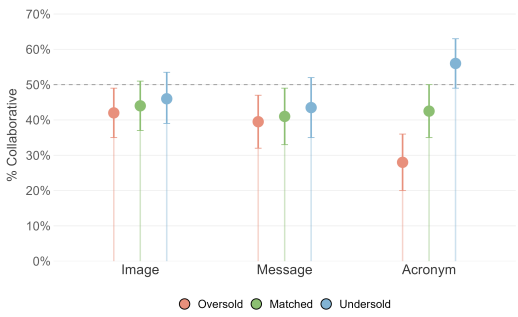


Figure 5: Percentage of user messages classified as Collaborative (vs. Directive) by task and framing condition. Error bars are 95% CIs. Dashed line at 50% marks an equally directive/collaborative split.

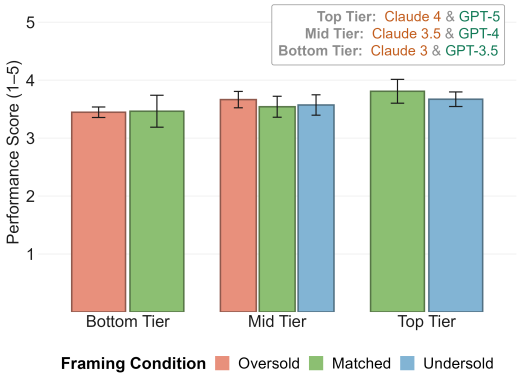


Figure 6: Mean task performance by source model tier and framing condition, with 95% CIs. There are seven bars rather than nine because a bottom-tier model cannot be Undersold and a top-tier model cannot be Oversold, as no lower or higher tier exists to show (Table 1).

Self-efficacy change and performance were weakly correlated (r=0.21, p=0.008; Appendix Figure 19): participants who scored higher gained slightly more self-efficacy, but the relationship was loose and the framing conditions overlapped heavily on performance. How capable participants felt and how well they actually performed therefore tracked each other only loosely.

3.4 Impression change is driven by experience, not output

We ran bivariate OLS regressions of Δ_{UTAUT} and Δ_{Godspeed} on three candidate predictors: objective task performance, self-efficacy change (Δ_{SE}), and expectations met. All predictors and outcomes were z-scored, so coefficients are standardized effect sizes. Results are in Table 4.

How well participants performed did not predict how their impression changed. The effect on Δ_{UTAUT} was essentially zero ($\beta = -0.01$, p=0.938; Figure 8), and the effect on Δ_{Godspeed}, though larger, was still not significant (β=0.11, p=0.180; Appendix Figure 25).

Self-efficacy change was a strong predictor: participants who finished the study feeling more capable as collaborators rated the model more favorably, and a one standard-deviation gain in self-efficacy corresponded to a 0.47 standard-deviation rise in Δ_{UTAUT} ($p < .001$; Figure 9) and a 0.36 standard-deviation rise in Δ_{Godspeed} ($p < .001$; Appendix Figure 26). This aligns with evidence that active AI collaboration preserves users' sense of competence better than passive reliance does (Lee et al., 2026).

Whether the model met participants' expectations had a comparably strong effect on impression change. Those who felt the model fell short revised their impression downward; those who felt it exceeded revised upward. A one standard-deviation increase in expectations met corresponded to a 0.47 standard-deviation gain in Δ_{UTAUT} and a 0.50 standard-deviation gain in Δ_{Godspeed} (both $p < .001$; Appendix Figures 27 and 28).

Expectations met on each of the three tasks predicted impression change, most strongly on the

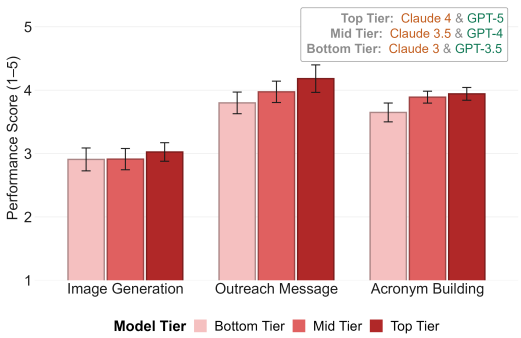


Figure 7: Mean task performance by task and source model tier, with 95% CIs.

Predictor	Outcome	β	p
Performance	Δ UTAUT	−0.01	0.938
	Δ Godspeed	0.11	0.180
Δ Self-Efficacy	Δ UTAUT	**0.47*****	$<.001$
	Δ Godspeed	**0.36*****	$<.001$
Expectations met	Δ UTAUT	**0.47*****	$<.001$
	Δ Godspeed	**0.50*****	$<.001$

Table 4: Bivariate OLS regressions of impression change on each candidate predictor (N=162 per regression). Predictors and outcomes are independently z-scored, so β is fully standardized. Asterisks on β: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

final acronym task and least on outreach, whose ratings were near-uniform across conditions and so left less variance to explain (Appendix Table 21).

Both experiential predictors tracked impression change, while objective performance did not. What moves users' ratings of a model is the experience of the interaction, not the output it produced.

4 Related Work

Framing Effects in Human-AI Interaction The presentation of AI systems shapes user perception and behavior independently of actual capability (Kosch et al., 2023; Grimes et al., 2021). Pre-interaction framing significantly alters baseline acceptance and judgments of system trustworthiness (Kocielnik et al., 2019; Pataranutaporn et al., 2023), and these expectation cues bias evaluations upward or downward independently of actual quality (Sun et al., 2026), influence reliance decisions (Cabrera et al., 2023), and shape active use as users develop implicit assumptions about underlying capabilities (Bangerl et al., 2025; Mohanty et al., 2025). Beliefs about capability drive reliance from both sides of the interaction: a model's stated accuracy shapes user trust alongside the accuracy users observe in practice (Yin et al., 2019), while overestimating one's own competence reduces appropriate reliance on the system (He et al., 2023). Our work builds on this by controlling the underlying model to create explicit expectation violations and examining how capability framing shifts user impressions of their assigned model.

Subjective versus Objective LLM Evaluation Standard LLM evaluations measure true capability through automated, model-focused benchmarks (Kazemi et al., 2025; Zhong et al., 2024; Hendrycks et al., 2021; Liang et al., 2023), yet these metrics often fail to capture the contextual nuances of how users evaluate models in practice (McIntosh et al., 2026; Sheikhi et al., 2026). Human-centric frameworks instead capture subjective experiences through in-the-moment satisfaction (Liu et al., 2025) and collaborative self-efficacy (Ju et al., 2025), but subjective perceptions frequently diverge from objective reality: evaluators rely on flawed linguistic heuristics (Jakesch et al., 2023), miscalibrate confidence (Steyvers et al., 2025), and rate AI-labeled content lower regardless of quality (Zhu et al., 2025). Contrasting subjective self-efficacy with objective output quality allows us to directly assess which more strongly drives changes in user impressions.

Interaction Behavior with AI Models Human interaction with LLMs ranges from directive commands to collaborative partnering (Li et al., 2024; Thu and Kocaballi, 2025). Researchers typically operationalize these dynamics through structural metrics like query adjustments and turn patterns (Liang et al., 2025; Wang et al., 2024; Koyuturk et al., 2025), which are shaped more by a system's presentation than its underlying architecture (Kraishan, 2026). Treating capability framing as the manipulated variable lets us observe how prompting behavior shifts while output quality stays fixed.

5 Discussion and Conclusion

Framing effects appear in our data where prior work predicts them, in attitudes and in reliance, and go no further: the quality of the artifact is unaffected. The difference is in what the task asks of the user. In advice-taking studies the participant decides whether to accept the system's recommendation, and that decision is both a measure of trust and the thing that determines whether the answer is correct (Yin et al., 2019; He et al., 2023), so an effect on trust becomes an effect on performance

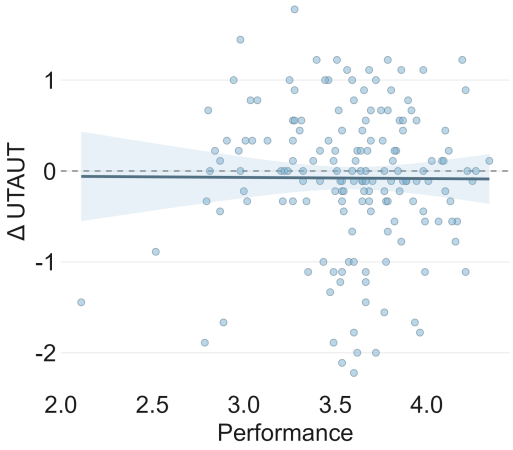


Figure 8: UTAUT change as a function of overall performance. OLS fit; 95% CI band, N=162.

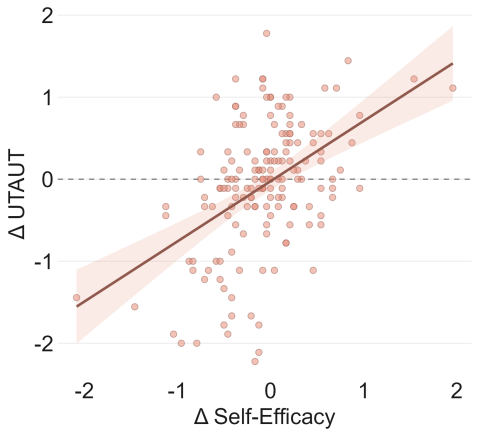


Figure 9: UTAUT change as a function of overall self-efficacy change (Δ_{SE}). OLS fit; 95% CI band, N=162.

almost by definition. In sustained collaboration the model produces the output no matter how the user prompts it, so trust and result can move independently.

A user's evaluation of an LLM is not a clean readout of its capability. For individual users, this means head-to-head self-reports between models of unequal reputation should be read with care: such reports encode the model's reputation alongside its actual performance (Sun et al., 2026; Pfeuffer, 2026). For organizations deploying AI internally, the implication is more actionable: how a tool is introduced may matter as much as which tool was chosen. Overselling to drive adoption risks depressing post-deployment satisfaction; setting realistic expectations may encourage more collaborative engagement with the tool (Davis, 1989; Übellacker, 2025).

Because impression change after use is driven by whether expectations were met rather than by what users produced, the way capability is communicated to users matters distinctly from the capability itself. For developers and benchmark designers, leaderboard rankings function as expectation-setting commitments (Bean et al., 2026), but the capability differences these rankings advertise are less perceptible to users during use than the rankings imply. A model marketed honestly is more likely to land in users' *exceeded expectations* bucket than one marketed at its benchmark ceiling.

For evaluation research, user-preference data that drives pipelines like Chatbot Arena (Chiang et al., 2024) partly measures expectation management (Wu and Aji, 2025; Hosking et al., 2024). Without controls that hold expectations constant across compared models, such data should be read as *experience relative to expectation*, not capability in isolation. Any evaluation routed through user judgment therefore needs identity blinding or a measured-expectation covariate, whether it is preference data, an in-product rating, or a satisfaction survey. Artifacts co-produced with users are more robust: participants' priors threaten the ratings attached to those artifacts, not the work itself.

LLM capability is real, and benchmarks measure something. But what users walk away believing is shaped at least as much by what they were told about a model as by what it did. The model and the message about it are not separate things to a user; they arrive together, and they leave together.

Future work. The larger question is how user-facing evaluation can be built to survive the priors users bring to it. Expectation goes unmeasured in the pipelines that collect user judgments today, and a short instrument administered alongside a rating would let it be reported or controlled for rather than absorbed into the score. What blinding buys is also open, since an anonymous protocol removes a model's identity only when users cannot recover it from the output. The same holds for automated evaluation, where it remains less clear whether naming the model behind an output shifts a judge's scores as our landing page shifted participants'. And because our effects concentrated on the most open-ended task, evaluation suites could be triaged by how much room each component leaves for expectation to work. Extensions of the study itself are noted in Limitations.

Limitations

Population. Our sample was US-based, English-speaking, and drawn from Prolific with a high prior approval rate. Crowdsourced workers may differ from casual consumer users or workplace deployments in ways that could amplify or attenuate framing effects; replication across more naturalistic samples would test the generalizability of these effects. As well, sensitivity to framing may also vary with cultural context, which a cross-locale replication would test.

Sample sizing. The N=162 target was derived from a Monte Carlo power analysis seeded with a pilot (n=18). However, effect estimates from n=18 are themselves noisy; the final sample is therefore sized against a deliberately conservative effect estimate rather than a precise one.

Models and tier ordering. Tier labels were anchored to public benchmark rankings at the time of data collection. The absolute capability gap between adjacent tiers is not equal across families, and the two bottom-tier models are distinct systems with different stylistic tendencies. We used two model families specifically to test the manipulation's robustness across different lineages; the families differ modestly in overall score (Appendix Table 22), and the central results are unchanged with family as a fixed effect. A finer-grained mapping between perceived capability and benchmark position would strengthen future replications. Our background measures also capture general familiarity with chatbots rather than preconceptions about the specific models named on the landing page, so model-specific reputations remain untested.

The manipulation also depends on model names whose tier ordering a general participant pool already recognizes, which is why we used the GPT and Claude lineages. Open-weight families would test whether framing effects survive where that ordering is less publicly established.

Task order. Tasks were presented in a fixed order: image generation, outreach message, then acronym building. The acronym task is also the most open-ended of the three, so the larger framing effect observed there is consistent with two non-mutually-exclusive accounts: (a) framing effects concentrate on tasks where user input most shapes what counts as a good output, and (b) framing effects compound over session time as users accumulate experience filtered through their initial expectation. Our design cannot distinguish these. A counterbalanced replication would separate task openness from time-in-session and would also test whether framing effects strengthen, attenuate, or reverse across repeated exposures within a session.

LLM judges and family-specific agreement. Task performance was rubric-scored by GPT-5 and message style was classified by `claude-haiku-4-5`. We averaged three independent scoring runs per output and validated the judge against three independent human raters on a subsample of submissions (Appendix A.7). Agreement among the human raters was modest, reflecting the inherent subjectivity of the rubrics, particularly for humor and persuasiveness. Computed on the same statistic, the judge agreed with the raters about as closely as the raters agreed with one another (ρ=0.301 and 0.298). Agreement was notably stronger for Claude-family outputs (ρ=0.483) than for GPT-family outputs (ρ=0.179, n.s.). Our performance results in §3.3 and in §3.4 should therefore be read as more reliable on the Claude side, although the central dissociation (framing affects behavior and impressions but not output) does not depend on the absolute calibration of the judge.

Similarly, the message classifier, run three times per message with a majority vote, was validated in the same way against three human annotators on a stratified sample of 100 messages (Appendix A.8). Agreement among the annotators was moderate (Fleiss κ=0.506), reflecting how interpretive the collaborative/directive distinction is at the message level, and the classifier reached fair agreement against that ceiling (Cohen κ=0.269). Because the classifier was blind to framing condition, its errors are distributed identically across groups and attenuate group differences rather than creating them, so the collaboration gradient in §3.2 persisted despite this noise rather than because of it.

Cross-sectional design. Impression change was measured within a single session, which is appropriate for studying how framing shapes an initial impression but does not address how those effects evolve with repeated exposure. Whether framing-induced impressions persist, erode, or compound over time is a question for longitudinal designs.

Statistical testing. We report uncorrected p-values. Our safeguard against false positives is built into the design rather than into a correction: every

effect of interest replicates across the two independent impression measures (UTAUT and Godspeed), and in §3.4 across two independent experiential predictors (self-efficacy change and expectations met). This cross-measure consistency provides design-level robustness for the central effects.

Predictor independence. Expectations-met and impression change are related but distinct: the former is rated per-task on a three-point scale immediately after each task; the latter is the difference between session-level composites on two multi-item validated scales. The two are measured on different scales, at different time points, and against different referents, so the strong association between them is informative rather than mechanical. Self-efficacy change, measured against the user's own capability rather than the model's, provides a fully independent experiential predictor; that both track impression change while objective performance does not is the pattern of interest.

Ethics Statement

All study procedures were approved by the authors' institutional research ethics board prior to data collection. Participants were recruited through Prolific and provided informed consent before any study activities. They were compensated at a rate of $12.50 USD per hour to be consistent with Prolific's recommended fair-pay guidelines, with an additional $10 performance bonus available on each of the tasks. No personally identifying information was collected; participant records were keyed only by Prolific worker ID, which was discarded after compensation processing. Conversation logs and survey responses were stored on access-controlled infrastructure. Participants were informed in the consent form that they would be interacting with a large language model and that their interactions would be analyzed; the specific framing manipulation was disclosed in a debrief at the end of the study.

The framing manipulation involved presenting some participants with a model description that did not match the model they actually used. This deception was minimal, time-limited, fully reversed in the debrief, and judged by the ethics review board to pose no risk of harm. We see no reasonably foreseeable harms from the methods or findings of this work; if anything, the findings argue for more transparent communication of AI system capability to users.

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A Appendix

This appendix collects the survey instruments, task materials, evaluation rubrics, example outputs, and supplementary analyses referenced in the main paper.

A.1 Survey Instruments

This section reproduces the items used in each scale, describes how each was administered, and describes how composite scores were computed.

Administration. The UTAUT performance expectancy and effort expectancy subscales, along with one additional engagement item, were administered in future tense before the study and in past tense after the study, with one embedded attention check per administration (Table 5). The Godspeed Perceived Intelligence subscale (Table 7) used the same five bipolar adjective items both pre- and post-study, as well as one additional bipolar adjective of "Artificial vs Natural". The task-specific NGSE items (Table 6) were administered before each of the three tasks in future tense and after each task in past tense.

UTAUT composites. Each administration yields a single score: the arithmetic mean of all non-attention items on a 1–5 Likert scale. Pre-study UTAUT is the mean across the pre-study items (Table 5, left column); post-study UTAUT is the mean across the post-study items (right column). Δ_{UTAUT} is post-study minus pre-study.

Godspeed Perceived Intelligence composites. The six bipolar items (Table 7) are each rated on a 1–7 semantic differential scale and averaged. The same items appear pre- and post-study; $\Delta_{\mathrm{Godspeed}}$ is post-study minus pre-study.

Task-specific NGSE composites. Each task has eight task-specific items (Table 6), each rated on a 1–5 Likert scale. The pre-task and post-task scores for a given task are the mean across non-attention items. Per-task self-efficacy change is post-task minus pre-task. Overall self-efficacy change (Δ_{SE}) is the unweighted mean of the three per-task changes.

Expectations met. After each task, participants chose one of three options describing whether the model fell short of, met, or exceeded their expectations for that task. The three options were coded as 1, 2, and 3 respectively. The composite *expectations-met* score used in §3.4 is the average of the three per-task codings, ranging from 1 (fell short on all three) to 3 (exceeded on all three).

A.2 Participant Exclusion Criteria

Every conversation log was reviewed manually after data collection and before any outcome analysis, blind to framing condition.

Insufficient engagement. Participants were told that each task required at least five minutes of interaction. A participant was removed if at least two of their three logs showed conversations substantially shorter than five minutes, or containing only one or two prompts. Participants who fell short on a single task but engaged substantially on the other two were retained.

Non-adherence. Non-adherence meant conversations wholly unrelated to the task, such as asking the model unrelated trivia questions, generating images unrelated to the target, or inventing new letter sets on the acronym task.

Counts. Ten participants were removed, six for non-adherence and four for insufficient engagement. Each was replaced through new recruitment, preserving nine participants per source-by-shown cell and the balanced $N{=}162$.

Perceived Intelligence
Incompetent – Competent
Ignorant – Knowledgeable
Irresponsible – Responsible
Unintelligent – Intelligent
Foolish – Sensible
Additional Pair
Artificial – Natural

Table 7: Godspeed Perceived Intelligence items and additional bipolar pair, each rated on a semantic differential scale from 1–7, identical across pre- and post-study administrations.

A.3 Task Materials

This section reproduces the materials participants saw: the capability framing landing page (Figure 10), the chat interface they used to complete all three tasks (Figure 11), the fixed target image for the image generation task (Figure 14), and the job description used in the outreach message task (Table 8).

Pre-study (future tense)	Post-study (past tense)
Performance Expectancy	
1. I think that I will find this chatbot useful	1. I found this chatbot useful
2. I expect that using this chatbot will enable me to accomplish tasks more quickly	2. Using this chatbot enabled me to accomplish tasks more quickly
3. I expect that using this chatbot will increase my productivity	3. Using this chatbot increased my productivity
4. If I use this chatbot, I feel I will increase my chances of getting my work done successfully	4. Using this chatbot, I felt my chances of getting my work done successfully was increased
Effort Expectancy	
5. I think I will find this chatbot easy to use	5. Please select "Disagree" for this item.
6. I think my interactions with this chatbot will be clear and understandable	6. I found this chatbot easy to use
7. I feel that it will be easy for me to become skillful at using this chatbot	7. My interactions with this chatbot were clear and understandable
8. Select "Strongly Agree" for this statement.	8. It was easy for me to become skillful at using this chatbot
9. I expect that learning to operate this chatbot will be easy for me	9. Learning to operate this chatbot was easy for me
Additional Engagement Item	
10. I plan to use the chatbot as much as possible during the tasks	10. I used the chatbot as much as possible during the tasks

Table 5: UTAUT Performance Expectancy and Effort Expectancy items with an additional engagement item added to the composite, each rated on a 1–5 Likert scale. Cells highlighted in *red* are attention checks; these were excluded from all subscale composites.

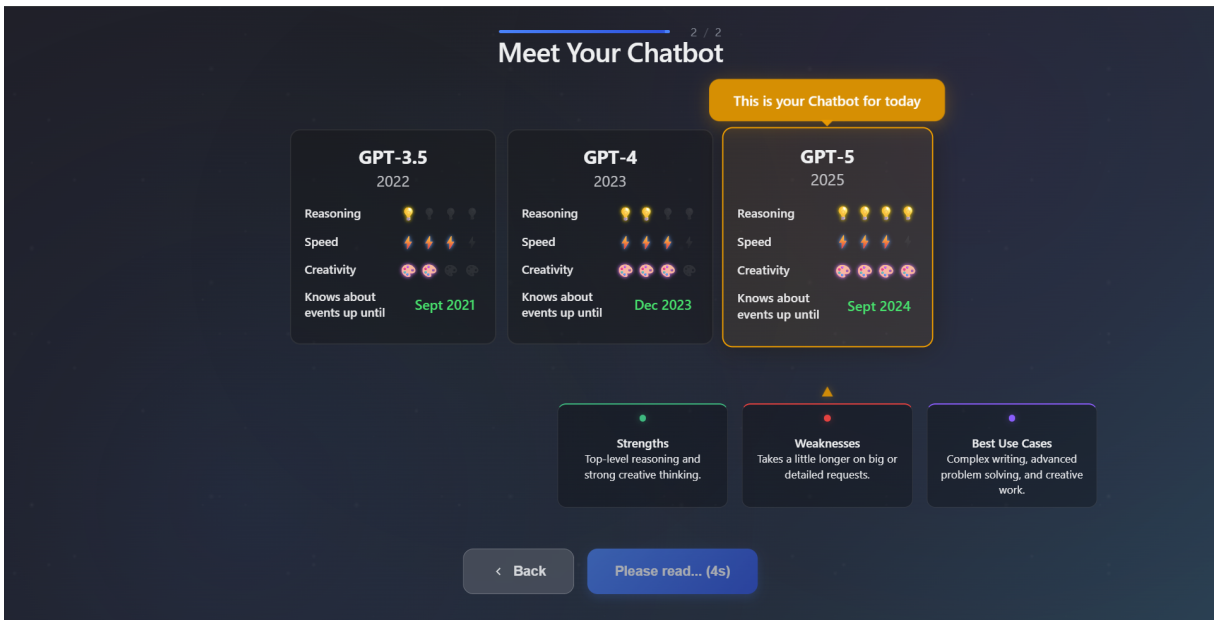


Figure 10: The landing page that constituted the capability framing manipulation.

Pre-task (future tense)	Post-task (past tense)
Image generation task	
1. I will be able to generate an image closest to the target image using the chatbot.	1. I was able to generate an image closest to the target image using the chatbot.
2. When facing difficulty with the image generation task, I am certain I will still accomplish it with the chatbot.	2. When facing difficulty with the image generation task, I was certain I would still accomplish it with the chatbot.
3. In general, I think that I can obtain outcomes that are important to achieving the target image using the chatbot.	3. In general, I thought that I obtained outcomes that were important to achieving the target image using the chatbot.
4. I believe I can succeed at the image generation task using the chatbot.	4. I believe I succeeded at the image generation task using the chatbot.
5. I will be able to successfully overcome challenges in the image generation task using the chatbot.	5. I was able to successfully overcome challenges in the image generation task using the chatbot.
6. Using the chatbot, I am confident that I can perform the image generation task effectively.	6. Using the chatbot, I was confident that I performed the image generation task effectively.
7. Compared to other people, I can do the image generation task very well with the chatbot.	7. Compared to other people, I think I did the image generation task very well with the chatbot.
8. Even when generating the target image is tough, I can perform quite well with the chatbot.	8. Even when generating the target image was tough, I performed quite well with the chatbot.
Outreach message writing task	
1. I will be able to write the most convincing message using the chatbot.	1. I was able to write the most convincing message using the chatbot.
2. When facing difficulty with the message writing task, I am certain I will still accomplish it with the chatbot.	2. When facing difficulty with the message writing task, I was certain I would still accomplish it with the chatbot.
3. In general, I think that I can obtain outcomes that are important to writing a convincing message using the chatbot.	3. In general, I thought that I obtained outcomes that were important to writing a convincing message using the chatbot.
4. Even if you don't agree, select Neutral below.	4. I believe I succeeded at the message writing task using the chatbot.
5. I believe I can succeed at the message writing task using the chatbot.	5. I was able to successfully overcome challenges in the message writing task using the chatbot.
6. I will be able to successfully overcome challenges in the message writing task using the chatbot.	6. Using the chatbot, I was confident that I performed the message writing task effectively.
7. Using the chatbot, I am confident that I can perform the message writing task effectively.	7. Compared to other people, I think I did the message writing task very well with the chatbot.
8. Compared to other people, I can do the message writing task very well with the chatbot.	8. Even when writing the message was tough, I performed quite well with the chatbot.
9. Even when writing the message is tough, I can perform quite well with the chatbot.	
Acronym building task	
1. I will be able to create the funniest acronyms using the chatbot.	1. I was able to create the funniest acronyms using the chatbot.
2. When facing difficulty with the acronym creation task, I am certain I will still accomplish it with the chatbot.	2. When facing difficulty with the acronym creation task, I was certain I would still accomplish it with the chatbot.
3. In general, I think that I can obtain outcomes that are important to creating a funny acronym using the chatbot.	3. In general, I thought that I obtained outcomes that were important to creating a funny acronym using the chatbot.
4. I believe I can succeed at the acronym creation task using the chatbot.	4. I believe I succeeded at the acronym creation task using the chatbot.
5. I will be able to successfully overcome challenges in the acronym creation task using the chatbot.	5. I was able to successfully overcome challenges in the acronym creation task using the chatbot.
6. Using the chatbot, I am confident that I can perform the acronym creation task effectively.	6. Using the chatbot, I was confident that I performed the acronym creation task effectively.
7. Compared to other people, I can do the acronym creation task very well with the chatbot.	7. Compared to other people, I think I did the acronym creation task very well with the chatbot.
8. Even when the acronym creation is tough, I can perform quite well with the chatbot.	8. Even when the acronym creation was tough, I performed quite well with the chatbot.

Table 6: Task-specific NGSE items for the three study tasks, rated on a 1–5 Likert scale. Cells highlighted in *red* are attention checks; these were excluded from all subscale composites.

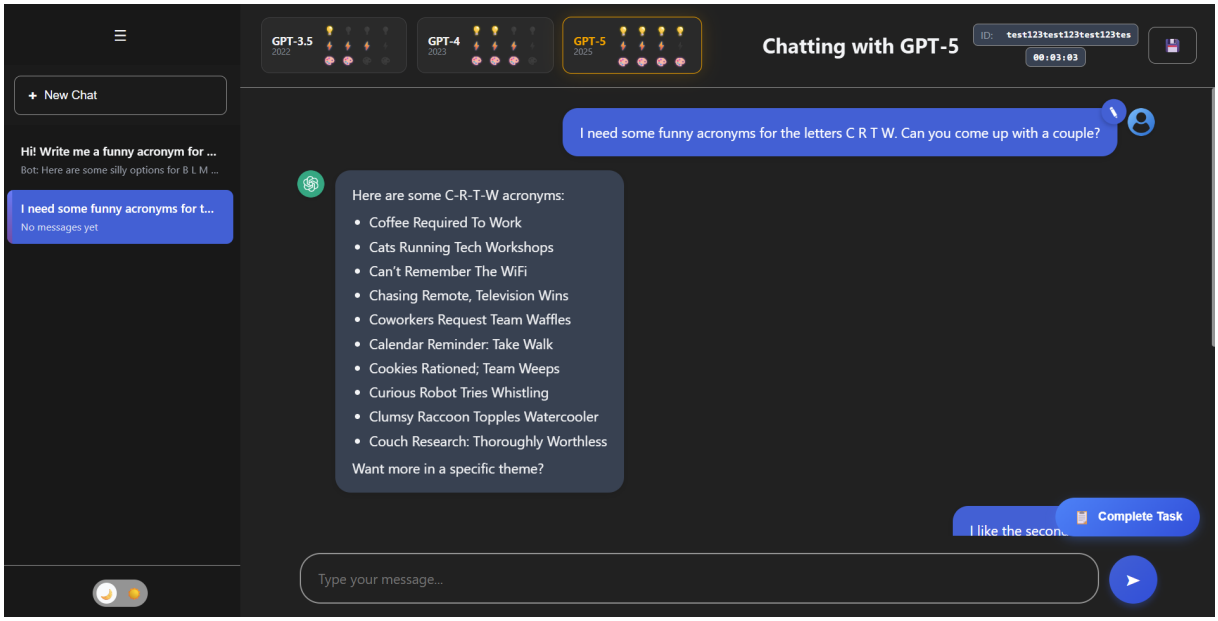


Figure 11: The chat interface during the acronym task.

Sales Representative – Dr. Rogers' Premier Office Furniture
We're seeking an energetic sales representative to join our established furniture dealership serving businesses across the region. You'll build relationships with office managers, architects, and facility planners to provide complete workspace solutions.
Key responsibilities:

- Generate new business through cold calling, networking, and referrals
- Conduct site visits and present furniture solutions to potential clients
- Prepare quotes, negotiate contracts, and manage the sales process
- Maintain relationships with existing accounts and identify growth opportunities.

Requirements:

- 2–4 years sales experience (B2B preferred, but not required)
- Strong interpersonal and presentation skills
- Valid driver's license and reliable transportation
- High school diploma required, college degree preferred.

What we offer:
Base salary ($40K) plus commission (avg. total $65K–80K), company car allowance, health benefits, and 401K matching in a stable, family-owned business.

Ready to build your sales career with us?

Table 8: The job description shown to participants for the outreach message writing task.

Criterion	Scale
Image Generation	
Accuracy. The generated image contains the same key objects, subjects, and elements as the target.	1 = Most key elements missing or wrong 5 = All key elements present and correct
Similarity. Layout, arrangement, and spatial relationships between elements match the target.	1 = Completely different arrangement 5 = Nearly identical layout and positioning
Style. Overall visual feel (color palette, lighting, atmosphere) matches the target.	1 = Completely different feel 5 = Very similar visual style and mood
Overall Resemblance. How similar the generated image is to the target overall.	1 = Not similar at all 5 = Very close match
Outreach Message Writing	
Naturalness. The message feels like it was written by a real person rather than an AI.	1 = Definitely AI-sounding 5 = Definitely human-sounding
Professionalism. The tone and style are appropriate for professional communication with a recruiter.	1 = Very unprofessional (casual, rude, etc.) 5 = Very professional (clear, respectful, polished)
Persuasiveness. How likely the reader would respond or be interested in the candidate.	1 = Not persuasive at all 5 = Very persuasive and credible
Quality. The message is written well and coherently.	1 = Written very poorly 5 = Written very well
Acronym Building	
Amusement. How funny the acronym is.	1 = Not funny at all 5 = Extremely funny
Novelty. The acronym shows a clever or novel twist; not predictable.	1 = Very predictable 5 = Very clever or unexpected
Coherence. The expanded phrase makes sense and follows the given letters correctly.	1 = Does not make sense / wrong letter order 5 = Makes sense and fits letters perfectly
Benignness. The acronym is appropriate; not offensive or harmful.	1 = Offensive / inappropriate 5 = Completely appropriate

Table 9: Rubric dimensions used by the LLM judge (GPT-5) for each task, scored on a 1–5 scale. The acronym rubric was applied separately to each of the three letter sets.

High-scoring outreach message (rubric mean 5.00)
Dear Dr. Rogers, I'm ready to step in as your next Sales Representative and drive growth for your established furniture dealership serving businesses across the region. In my previous territory, I generated new business through cold calling, networking, and referrals, averaging 45 targeted calls a day and 8–10 first meetings a week, which powered six straight quarters at 120%+ to quota and $1.1M TTM. I conducted site visits and presented furniture solutions to office managers, architects, and facility planners, turning walk-throughs into tailored proposals that raised our win rate from 22% to 33% and shortened the sales cycle by two weeks. I prepared quotes, negotiated contracts, and managed the sales process end to end, with a $22K average deal size and on-time installs. After the sale, I maintained relationships with existing accounts and identified growth opportunities: 95% retention, 35% average expansion within six months, and 30% of revenue from referrals. Give me 90 days to map the territory, fill the pipeline, and close early wins while setting up steady expansion. I'd welcome a conversation on how I can help grow your market share. Best regards, J. Doe
Low-scoring outreach message (rubric mean 2.33)
Dear Hiring Manager, I am excited to apply for the Sales Representative position at Dr. Rogers' Premier Office Furniture. In my previous roles, I have successfully generated new business through cold calling, networking, and referrals. I excel in conducting site visits and presenting tailored furniture solutions to potential clients. My experience includes preparing quotes, negotiating contracts, and managing the sales process efficiently. I have a proven track record of maintaining strong relationships with existing accounts and identifying growth opportunities. I am confident that my skills and experiences make me a strong fit for this role and I am eager to contribute to the success of your team. Thank you for considering my application. I look forward to discussing how I can bring value to Dr. Rogers' Premier Office Furniture. Warm regards, [Your Name]

Table 10: High- and low-scoring submissions from the outreach message task.



Figure 12: High-scoring participant submission for the image generation task (mean rubric score 4.33).



Figure 13: Low-scoring participant submission for the image generation task (mean rubric score 1.5).



Figure 14: The fixed target image shown to all participants for the image generation task.

A.4 Evaluation Rubrics and Example Outputs

Submitted outputs were scored by an LLM judge (GPT-5) against four rubric dimensions per task (Table 9). To illustrate the spread of submissions on each task, Figures 12 and 13 show a high- and low-scoring image submission. Tables 10 and 11 give high- and low-scoring outreach and acronym submissions.

High-scoring acronyms (rubric mean 4.22)	
BLMPF	Book Lovers Mourn Poor Films
CRTW	Cats Request Tuna Weekends
YOLD	Your Orientation Lacks Direction
Low-scoring acronyms (rubric mean 2.83)	
BLMPF	Boat Lamp Milk Pen Flower
CRTW	Cat Rabbit Turtle Watermelon
YOLD	Yacht Owl Lemon Dog

Table 11: High- and low-scoring acronym submissions across the three letter sets.

A.5 Collaborative/Directive Classification

Two models serve the two evaluation roles, chosen for different requirements. The judge scores one final artifact per participant per task and must be robust enough to evaluate across various criteria, thus the strongest model available to us was used, GPT-5. The classifier labels every user message in every log, close to 10,000 calls on a simpler binary decision, and uses `claude-haiku-4-5`, which provided reliable classification at that throughput at a fraction of the cost of frontier models. Drawing the two from different vendors also avoids depending on a single provider.

For the message classification pipeline, we used **`claude-haiku-4-5`** to label each participant message as collaborative or directive. Messages were classified three times using the default temperature

of 1.0 and max tokens set to 512, then the majority classification was used. The preceding AI response was truncated to 400 characters to fit within the model's context window along with the user message and instructions.

The definition of **directive** that was given to the model was adapted from Searle (1976) who defined it as the following:

> "they are attempts (of varying degrees, and hence, more precisely, they are determinates of the determinable which includes attempting) by the speaker to get the hearer to do something. They may be very modest 'attempts' as when I invite you to do it or suggest that you do it, or they may be very fierce attempts as when I insist that you do it."

The definition of **collaboration** that was given to the model was adapted from Roschelle and Teasley (1995) who defined it as the following:

> "Collaboration is a coordinated, synchronous activity that is the result of a continued attempt to construct and maintain a shared conception of a problem."

These adaptations can be seen in the system prompt that was used below. The examples given were manually coded by a human from real chat log data, based on the definitions.

Collaboration/Directive Classifier (system prompt):

```
You are classifying user messages from
human-AI task conversations. For each
message you will receive the previous
AI response (if available) and the user
message to classify.

Classify each user message as exactly
one of:

DIRECTIVE (D)
A message whose primary purpose is to
get the model to perform an action,
without engaging with or building on
its previous response. It is
transactional in nature, treating the
model as a tool to execute rather than
a partner to engage with. There is no
attempt to construct shared context or
build on what came before.
Examples:
- "No",
- "Try again",
- "Something funnier",
- "Make it make sense",
- "Create a funny acronym for BLMPF"

COLLABORATIVE (C)
A message that works toward a mutual
construction of a shared conception of
the task, by building on, referencing,
or extending the model's previous
response, or by providing rich context
that frames the interaction as a joint
endeavour. It treats the model as a
partner rather than a tool,
contributing to a shared understanding
of what is being created together.
Examples:
- "The blobfish one is my favourite,
can you think of others involving
blobfish?",
- "I liked the animal theme, keep that
but make it funnier",
- "Can you help me create a humorous
acronym for BLMPF, I'd love it if it
was about an animal doing something
unusual"

DISAMBIGUATION
If both elements are present, classify
by primary communicative function.
Briefly acknowledging before commanding
("That was okay, try again") is
Directive. Substantively building on
the previous output ("I liked the
animal theme, can we keep that?") is
Collaborative.

For the first message in a
conversation, a bare command or simple
request is Directive. A message
providing creative context,
constraints, or framing is
Collaborative.

Return ONLY a JSON array, one object
per message, in the same order received:
[{"i":1,"label":"D"},{"i":2,"label":"C"},
...]
No other text.
```

Each turn was formatted as:

```
{i}.
AI: "{preceding AI response, up to 400
characters}"
User: "{user message}"
```

A.6 LLM Judge for Output Quality

We used **GPT-5** as a judge to score participant outputs against the rubrics for each task. Each output was scored in three independent runs, with scores averaged to produce a final point estimate, using a default temperature of 1.0 for all runs. The system prompt for the judge was varied by task to include the relevant rubric and instructions, but the user message below was consistent across tasks:

> "Please evaluate the following [task content] according to the rubric provided in

your instructions. Return ONLY a JSON object."

For the image generation task, the participant's generated image and the target image were included as base64-encoded PNG data.

Below are the system prompts that were given to the model when evaluating each task.

Image generation (system prompt):

```
We conducted a study where participants
were given a target image and asked to
prompt a model to generate an image as
close to the target as possible.

These were the instructions that they
were given:
"In this task, you will use the
chatbot's image generation capabilities
to create an image, trying to match a
target picture below as closely as
possible. The participant with the
closest image to the target will be
awarded a bonus of $10.
You can prompt the chatbot to generate
images using phrases like:
- "Generate a picture of..."
- "Make me an image of..."
As well, you can ask the chatbot to
add, remove, or adjust elements of the
image and it will generate a new version
The target image is below:
[target image]"

You will be given two images: the
participant's image first, then the
target image.

Evaluate the participant's image
against the target on the following 4
criteria, each scored 1-5:
q1.  Accuracy - The generated image
contains the same key objects,
subjects, and elements as the target
image.
- 1 = Most key elements are missing or
wrong
- 5 = All key elements are present and
correct

q2.  Similarity - The layout,
arrangement, and spatial relationships
between elements match the target image.
- 1 = Completely different arrangement
- 5 = Nearly identical layout and
positioning

q3.  Style - The overall visual feel of
the generated image (color palette,
lighting, atmosphere) matches the
target image.
- 1 = Completely different feel
- 5 = Very similar visual style and mood

q4.  Overall Resemblance - How similar
is this image to the target?
- 1 = Not similar at all
- 5 = Very close match

Return ONLY a JSON object in this exact
format:
{"q1": <score>, "q2": <score>, "q3":
<score>, "q4": <score>}
```

Outreach message (system prompt):

```
We conducted a study where participants
were asked to write an outreach message
to a hiring manager based on a job
description, alongside a chatbot model.

These were the instructions that they
were given:
"Below is a job description for a sales
representative posted by Dr. Rogers.
Working with the chatbot, your task is
to write a message to Dr. Rogers
explaining why you are the best
candidate for the job with the goal of
being hired.
The participant who creates the most
convincing message (most likely to be
hired) will receive a bonus of $10.
Restrictions:
- Must contain some direct elements
from the job description
- Must sign name as "J. Doe"
- Must address it to "Dr. Rogers"
- Should not sound AI-generated

Job Description below:
Sales Representative - Dr. Rogers'
Premier Office Furniture
We're seeking an energetic Sales
Representative to join our established
furniture dealership serving businesses
across the region. You'll build
relationships with office managers,
architects, and facility planners to
provide complete workspace solutions.
Key Responsibilities:
- Generate new business through cold
calling, networking, and referrals
- Conduct site visits and present
furniture solutions to potential clients
- Prepare quotes, negotiate contracts,
and manage the sales process
- Maintain relationships with existing
accounts and identify growth
opportunities
Requirements:
- 2-4 years sales experience (B2B
preferred, but not required)
- Strong interpersonal and presentation
skills
- Valid driver's license and reliable
transportation
- High school diploma required; college
degree preferred
What We Offer:
Base salary ($40K) plus commission
(avg. total $65K-$80K), company car
allowance, health benefits, and 401K
matching in a stable, family-owned
business.
Ready to build your sales career with
us?"
```

```
Evaluate the message on the following 4
criteria, each scored 1-5:
q1. Naturalness - This message feels
like it was written by a real person
rather than an AI.
- 1 = Definitely AI-Sounding
- 5 = Definitely Human-Sounding

q2. Professionalism - The tone and
style are appropriate for professional
communication with a recruiter.
- 1 = Very unprofessional (casual,
rude, etc.)
- 5 = Very professional (clear,
respectful, polished)

q3. Persuasiveness - How likely would
you respond to this person or be
interested in this candidate?
- 1 = Not persuasive at all
- 5 = Very persuasive and credible

q4. Quality - The message is written
well and coherently.
- 1 = Written very poorly
- 5 = Written very well

Return ONLY a JSON object in this exact
format:
{"q1": <score>, "q2": <score>, "q3":
<score>, "q4": <score>}
```

Acronym building (system prompt):

```
We conducted a study where participants
were given three sets of letters and
asked to make a funny acronym for each,
alongside a chatbot model.

These were the instructions that they
were given:
"In this task, you will be given a
series of letters and you will use the
chatbot to help build them into a funny
acronym. The participant with the
funniest acronym will be awarded a
bonus of $10.
For example, the following are letters
and possible acronyms that can be made
with them.
- PUGT = "Prolific Users Get Tired"
- LRP = "Laundry Remains Piled"
- USI = "Usually Sleeping In"
The acronym letters are below:
B L M P F
C R T W
Y O L D"

You will be given three acronyms.
Evaluate each one on the following 4
criteria, each scored 1-5:
q1. Amusement - How funny did you find
this acronym?
- 1 = Not funny at all
- 5 = Extremely funny

q2. Novelty - The acronym shows a
clever or novel twist.
- 1 = Very predictable
- 5 = Very clever or unexpected

q3. Coherence - The expanded phrase
makes sense and follows the given
letters correctly.
- 1 = Does not make sense / wrong order
- 5 = Makes sense and fits letters
perfectly

q4. Benignness - The acronym is
appropriate (not offensive or harmful).
- 1 = Offensive / inappropriate
- 5 = Completely appropriate

Return ONLY a JSON object in this exact
format:
{
  "acr1_q1": <score>, "acr1_q2":
<score>, "acr1_q3": <score>, "acr1_q4":
<score>,
  "acr2_q1": <score>, "acr2_q2":
<score>, "acr2_q3": <score>, "acr2_q4":
<score>,
  "acr3_q1": <score>, "acr3_q2":
<score>, "acr3_q3": <score>, "acr3_q4":
<score>
}
```

A.7 Human Validation of the LLM Judge

To check the LLM judge against an external standard and to test for systematic family-specific bias, we had three independent human raters score a sample of participant submissions for each task using the same rubrics as the LLM judge (Table 9).

Raters. Five members of our lab served as raters. Three were assigned to each task, so some raters covered the assessment of multiple tasks. They spanned undergraduate to faculty level, coming from Computer Science, Applied Mathematics and Statistics, and Business. All were aware of the general aims of the project, but none had seen the LLM judge's scores or any other study data before rating. All raters joined on a volunteer basis.

Blinding. Raters were blind to (i) the source model identity (model family and tier), (ii) the framing condition assigned to the participant, and (iii) the LLM judge's scores for the same submissions.

Sample. We rated 54 submissions per task. To set this size, we first ran a pilot in which raters scored 18 submissions, one from each study configuration (nine model-framing combinations per family, across both families). We then calculated the Intraclass Correlation Coefficient (ICC) from that pilot in a power analysis targeting 80% power, indicating that a sample of 54 participants would be sufficient. The final sample was drawn pro-

portionally from the full dataset, preserving the distribution of framing conditions.

Procedure. Rating was administered through Qualtrics. Each rater began with an introductory page describing the task and the rubric, then rated one submission per page on the four rubric dimensions on a 1–5 scale, moving freely between submissions as they went. For the image task, the participant's submission and the fixed target were shown side by side; for acronym building, the letter set and the participant's expansion were shown together; for outreach, however, only the participant's message was shown, since the rubric does not depend on the original job description. Figure 15 shows a sample of the rater's interface during evaluation of the acronym building task. Approximate rating times varied by task, with the image generation and acronym building tasks generally taking around a minute per submission, and the outreach message task taking between one and two minutes.

Inter-rater reliability. Inter-rater reliability was calculated using pairwise Spearman correlation. Agreement among the three raters was modest, with ρ of 0.387 for image generation, 0.280 for outreach message writing, and 0.226 for acronym building, shown in Table 12. However, these values reflect the subjectivity of some rubrics: image generation has a clear target to compare against, while acronym building asks raters to judge how personally humorous they found the submission, where reasonable people often disagree.

Agreement with the LLM judge. The LLM judge's scores correlated with the human consensus (the mean of the three raters) at ρ=0.575 for image generation, ρ=0.171 for outreach, and ρ=0.156 for acronym building (overall mean ρ=0.301; Table 12). This shows that overall, the model's evaluations were similar in agreement to the human evaluators, as the human evaluators were to each other. Human agreement is the practical ceiling for an automated scorer on rubrics this subjective, since a judge cannot reasonably be expected to agree with raters more than they agree among themselves. Agreement was highest for humans and judge alike on image generation, the one task with an external target, and fell for both on the two open-ended tasks, which points to task subjectivity rather than judge unreliability as the source of the lower correlations.

Task	**Human–human**	**Judge–human**
Image generation	0.387	0.575
Outreach message	0.280	0.171
Acronym building	0.226	0.156
Overall	0.298	0.301

Table 12: Agreement on the validation subsample, reported as Spearman ρ. Human–human is the mean pairwise correlation among the three raters; judge–human correlates the LLM judge against the rater consensus.

Family	Mean Judge	Mean Human	Gap
GPT	3.85	3.52	+0.33
Claude	3.56	3.40	+0.15

Table 13: Mean judge rating and human annotator rating for each model family and the gap difference between them. The judge and human annotators evaluated the same subsample which was selected at random.

Family-specific bias. To test for systematic bias when the LLM judge scored outputs across model families, we compared LLM-human agreement separately for submissions produced with GPT-family and Claude-family models. Agreement was ρ=0.179 (p=0.372) for GPT submissions and ρ=0.483 (p=0.011) for Claude submissions. Therefore, the judge's scores aligned more closely with the human consensus on Claude outputs than on GPT outputs. The performance results in §3.3 should be read with this asymmetry in mind.

On average, the judge rated GPT-family outputs 3.85 against the average human rating of 3.52 (on the same subsample), and Claude-family outputs 3.56 against the average human rating of 3.40, which can be seen in Table 13. The judge-human gap for GPT was $+0.33$ and for Claude was $+0.15$, meaning the offset difference between the LLM judge and human annotators between models was 0.18 points and does not reach significance (p=0.07). Additionally, rerunning the regression in §3.3 with a family fixed effect leaves the tier and framing coefficients *identical* to four decimal places with or without the family term (family fixed-effect tier $\beta = +0.4365$, p=0.0005; framing $\beta = +0.1357$, p=0.271). Therefore, any offset, if real, would have no bearing on the final results.

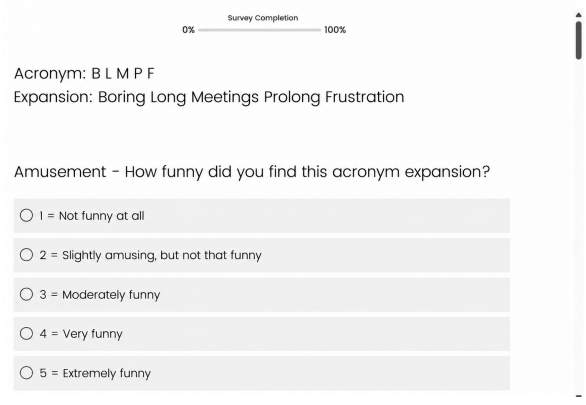


Figure 15: Qualtrics page for rating participant submissions on the acronym building task.

A.8 Human Validation of the Message Classifier

The collaborative/directive classifier was validated against human annotators in the same way as the LLM judge.

Sample. Three annotators labeled a stratified sample of 100 messages drawn from the full pool of 3171 classified messages, balanced across the three tasks (image 34, acronym 34, outreach 32) and across the two classifier labels, using the same definitions given to the classifier (Appendix A.5). Stratifying by task and label prevents the agreement estimate from being inflated by easy agreement on one dominant task or label.

Agreement. Agreement among the annotators was itself only moderate (Table 14), reflecting how interpretive the collaborative/directive distinction is at the message level, and the classifier reached fair agreement against that ceiling. Agreement was measured with Fleiss' κ among the three annotators and Cohen's κ for the classifier against the human consensus. By task, classifier agreement with the human consensus was κ=0.199 for image generation, 0.395 for outreach message writing, and 0.217 for acronym building.

Effect on the collaboration result. Two properties of the design keep this noise from threatening the gradient reported in §3.2. The classifier was blind to framing condition, so its errors are distributed identically across groups; misclassification of that kind attenuates group differences rather than creating them, and the gradient on the acronym task therefore persisted despite the noise rather than because of it. Each message was also classified three times with a majority vote, which reduces run-to-run variance.

Comparison	κ	Interpretation
Human vs. human (Fleiss)	0.506	Moderate
Classifier vs. consensus (Cohen)	0.269	Fair
Classifier vs. individual raters	0.25–0.30	Fair

Table 14: Agreement on the 100-message validation sample for the collaborative/directive classifier. Human consensus is the majority label across the three annotators.

A.9 Behavioral Metrics

Table 15 lists the full set of behavioral metrics extracted from each participant's conversation logs on each task.

Metric	Definition
Total messages	Total number of messages sent by the participant during the task.
Mean message length	Mean character count per participant message.
First message length	Character count of the participant's first message.
Total keystrokes	Total number of keydown events recorded while typing a message.
Keystrokes per message	Mean number of keystrokes per message sent.
Messages per conversation	Mean number of messages sent per conversation thread.
Mean conversation depth	Average point in a conversation thread when the participant switched away from it or finished the task.
Edit count	Number of times the participant edited a previously sent message.
Mean edit distance	Average number of differences between edited and original messages.
Total backspaces	Total number of backspace key presses recorded while typing a message.
Conversation count	Total number of conversation threads started by the participant.
Conversation switches	Number of times the participant switched between conversation threads.
Total clicks	Total number of mouse click events recorded during the session.
Total mouse movements	Count of sampled mouse movement events, recorded at a 1-in-10 sampling rate.
Session duration	Total elapsed time from task page load to task completion.
Total idle time	Total time the task page was out of focus (tab hidden).
Mean inter-message gap	Mean elapsed time in seconds between consecutive message submissions.

Table 15: The 17 behavioral metrics that were tracked during testing per participant per task.

A.10 Representative Chat Logs

Figures 16 and 17 illustrate the qualitative difference in interaction style discussed in Section 3.2. The Oversold participant sent short, directive prompts and re-prompted quickly after each output. The Undersold participant sent fewer, longer prompts with more deliberation between them.

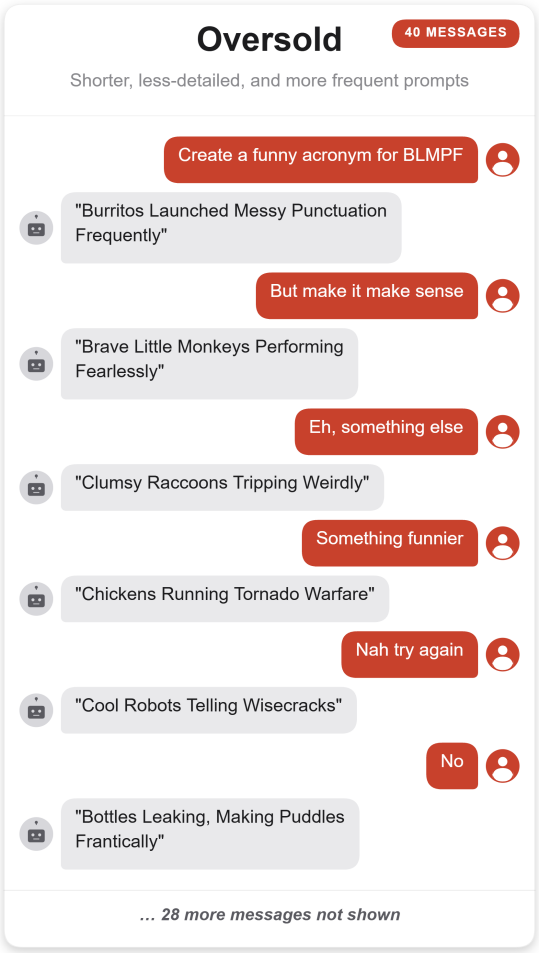


Figure 16: Representative chat log from an Oversold participant on the acronym task.

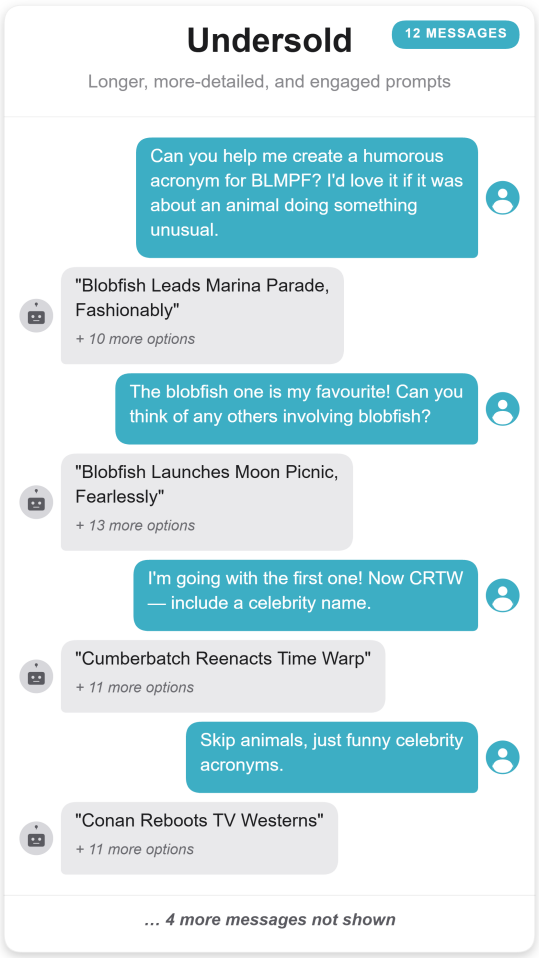


Figure 17: Representative chat log from an Undersold participant on the acronym task.

A.11 Supplementary Analyses

This section collects the supplementary analyses referenced in the Results. Table 16 reports the full pairwise grid of t-tests on pre-interaction perceptions, Figure 18 shows pre- and post-study score distributions pooled across all conditions, and Table 17 reports the framing $\times$ familiarity interaction tests, all referenced in Section 3.1. Table 18 reports the similarity between participants' submissions and the model's own text, and Table 19 reports per-task self-efficacy change and expectations met by framing condition, both referenced in Section 3.2. Table 20 reports two-way ANOVAs of performance on model tier and framing, pooled and per task, and Figure 19 shows the relationship between self-efficacy change and overall performance, both referenced in Section 3.3. Table 21 regresses overall impression change on each task's expectations-met rating, referenced in Section 3.4. Table 22 reports mean performance by model family, which is indistinguishable on image generation and outreach message writing and differs modestly on acronym building and overall. Figures 20–22 show performance by tier and framing condition broken down by task. Figures 23 and 24 replicate the acronym engagement metrics from the main paper on the image and outreach message tasks. Figures 25 and 26 show the Godspeed counterparts to the performance and self-efficacy scatter plots referenced in Section 3.4. Figures 27 and 28 show the scatter for the expectations-met predictor referenced in Section 3.4 for UTAUT and Godspeed respectively.

Outcome	Contrast	$t(106)$	p	d
Pre-UTAUT	Over vs. Matched	1.24	0.218	0.24
	Under vs. Matched	−0.73	0.467	−0.14
	Over vs. Under	**2.22***	0.029	0.43
Pre-Godspeed	Over vs. Matched	0.63	0.532	0.12
	Under vs. Matched	**−2.77****	0.007	−0.53
	Over vs. Under	**3.41*****	0.0009	0.66

Table 16: Pairwise independent-samples Student's t-tests on pre-interaction UTAUT and Godspeed by framing condition, equal variances assumed. d is Cohen's d. Asterisks on t: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

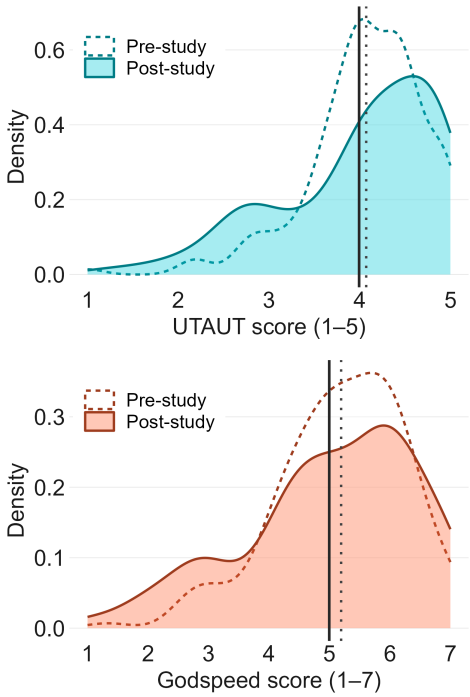


Figure 18: Pre- vs. post-study density of UTAUT (left) and Godspeed (right), pooled across conditions. Dashed outline: pre. Filled density: post. Dotted vertical: pre-study mean; solid vertical: post-study mean.

Familiarity measure	Outcome	Interaction β [95% CI]	p
Chatbot use frequency	Pre-UTAUT	+0.084 [−0.11, +0.28]	.393
Comfort with AI	Pre-UTAUT	+0.090 [−0.09, +0.27]	.316
Chatbot use frequency	Pre-Godspeed	−0.057 [−0.25, +0.14]	.565
Comfort with AI	Pre-Godspeed	−0.031 [−0.21, +0.15]	.737
Chatbot use frequency	ΔUTAUT	−0.106 [−0.30, +0.09]	.286
Comfort with AI	ΔUTAUT	−0.134 [−0.32, +0.05]	.154
Chatbot use frequency	ΔGodspeed	−0.094 [−0.29, +0.10]	.334
Comfort with AI	ΔGodspeed	−0.091 [−0.27, +0.09]	.326

Table 17: Framing $\times$ familiarity interaction terms from OLS models of pre-interaction impressions and impression change (N=162 per model). No interaction reaches significance.

Task	Over	Matched	Under	ANOVA p	Over vs. Under p
Outreach message	0.932	0.907	0.881	.353	.149
Acronym building	0.317	0.149	0.170	**<.0001**	**.001**

Table 18: Mean normalized Levenshtein similarity between each participant's final submission and the most similar assistant message in their conversation log, by task and framing condition (N=54 per condition). Acronym values average the three letter sets. Image generation is excluded because its submissions are produced by the model.

	Self-efficacy change				Expectations met			
Task	Over	Matched	Under	p	Over	Matched	Under	p
Image generation	−0.76	−0.66	−0.38	.100	1.50	1.56	1.69	.335
Outreach message	+0.41	+0.46	+0.26	.374	2.30	2.30	2.26	.933
Acronym building	−0.08	−0.18	+0.26	.052	2.06	1.98	2.37	**.013**

Table 19: Per-task self-efficacy change (post-task minus pre-task) and expectations-met rating (1 = fell short, 3 = exceeded) by framing condition, with one-way ANOVA p-values. N=54 per condition.

	Model tier		Framing	
Outcome	F	p	F	p
Overall	**6.12**	0.003	0.68	0.51
Image generation	0.65	0.52	1.20	0.30
Outreach message	**4.23**	0.016	0.86	0.42
Acronym building	**7.20**	0.001	0.94	0.39

Table 20: Two-way ANOVAs of task performance on source model tier and framing condition as categorical factors. The first row pools the three tasks; the remaining rows fit the same model within each task. Partial η^2 in the pooled model is 0.072 for tier and 0.009 for framing. Bold F values are significant at $p < 0.05$.

Task	ΔUTAUT β (p)	ΔGodspeed β (p)
Image generation	0.272 (<.001)	0.365 (<.001)
Outreach message	0.158 (.045)	0.190 (.015)
Acronym building	0.499 (<.001)	0.471 (<.001)

Table 21: OLS regressions of overall impression change on each task's expectations-met rating, fitted separately for each task (N=162 per regression). Predictors and outcomes are z-scored.

Task	GPT	Claude	t	p
Image generation	2.97	2.92	0.49	.628
Outreach message	4.04	3.94	0.91	.365
Acronym building	3.93	3.72	**3.08**	.002
Overall	3.76	3.61	**2.81**	.006

Table 22: Mean judge-scored performance by model family, with independent-samples t-tests per task and for the overall composite.

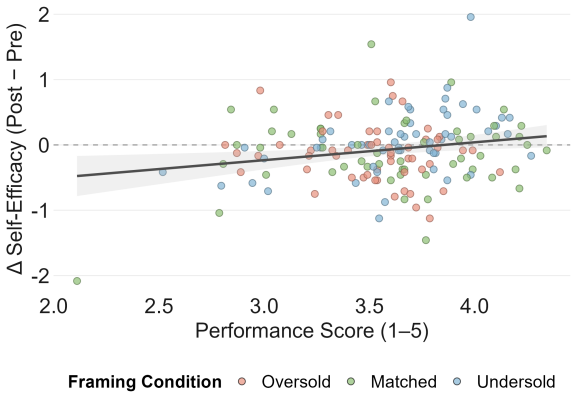


Figure 19: Overall self-efficacy change (Δ_{SE}) against overall performance, by framing condition. OLS fit pooled across conditions with 95% CI. Pearson r=0.21, p=0.008, N=162.

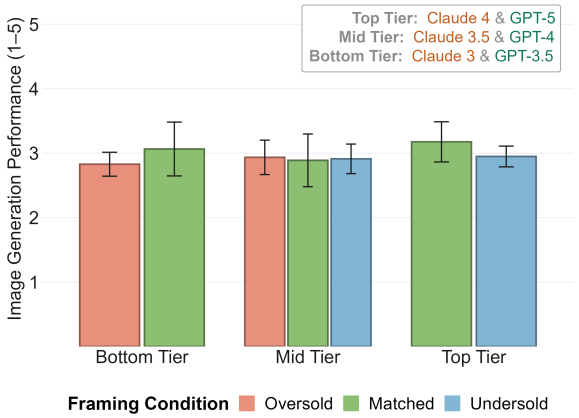


Figure 20: Mean task performance on the image generation task by source model tier and framing condition, with 95% CIs.

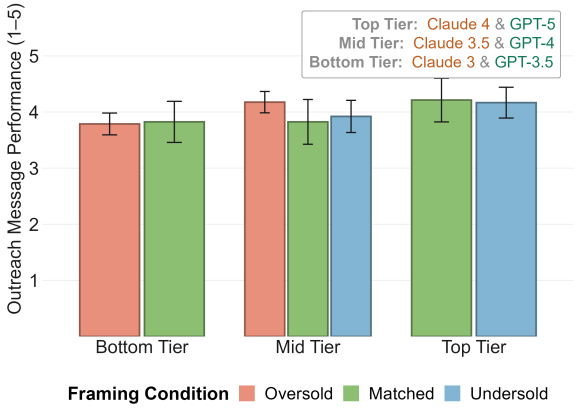


Figure 21: Mean task performance on the outreach message task by source model tier and framing condition, with 95% CIs.

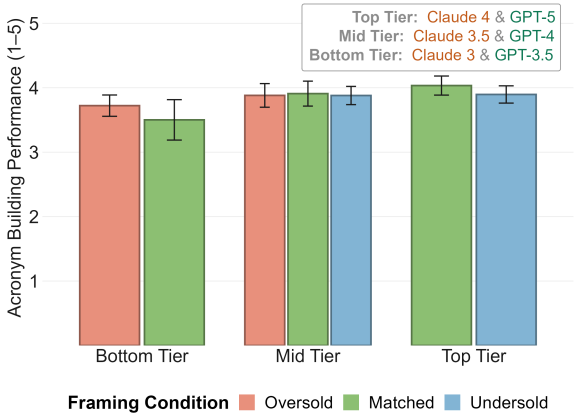


Figure 22: Mean task performance on the acronym building task by source model tier and framing condition, with 95% CIs.

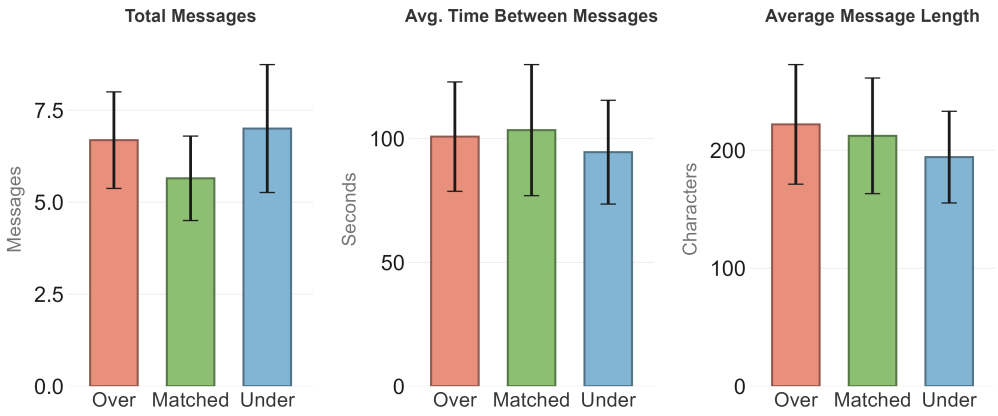


Figure 23: Three engagement metrics for the image generation task by framing condition (mean $\pm 95\%$ CI). From left to right: total messages, mean time between messages, mean message length.

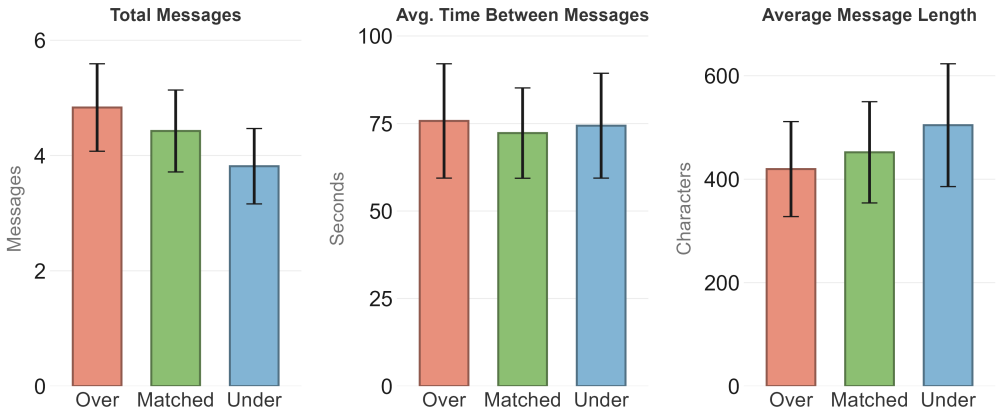


Figure 24: Three engagement metrics for the outreach message task by framing condition (mean $\pm 95\%$ CI). From left to right: total messages, mean time between messages, mean message length.

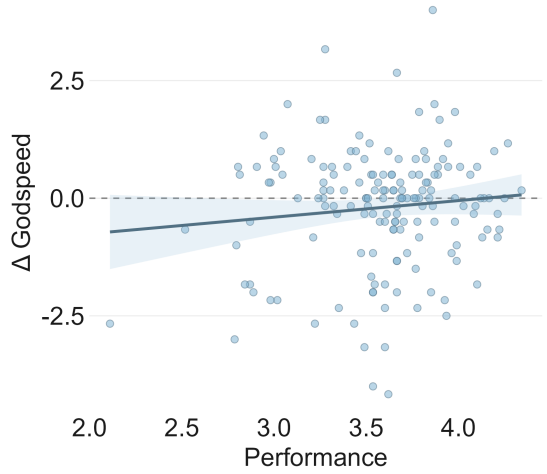


Figure 25: Godspeed change as a function of overall performance. OLS fit with 95% CI band, N=162.

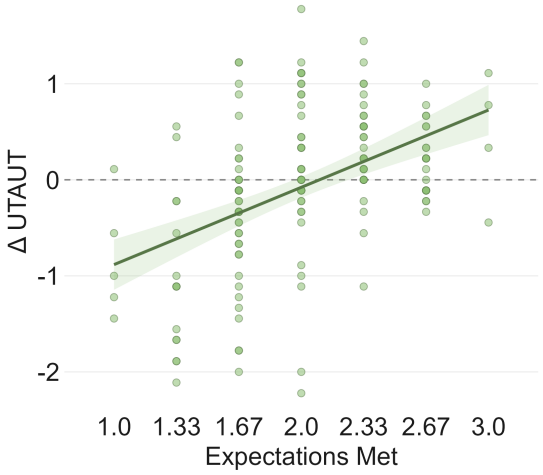


Figure 27: UTAUT change as a function of whether the model met task-level expectations (composite of three per-task ratings: fell short, met, exceeded). OLS fit with 95% CI band, N=162.



Figure 26: Godspeed change as a function of overall self-efficacy change (Δ_{SE}). OLS fit with 95% CI band, N=162.

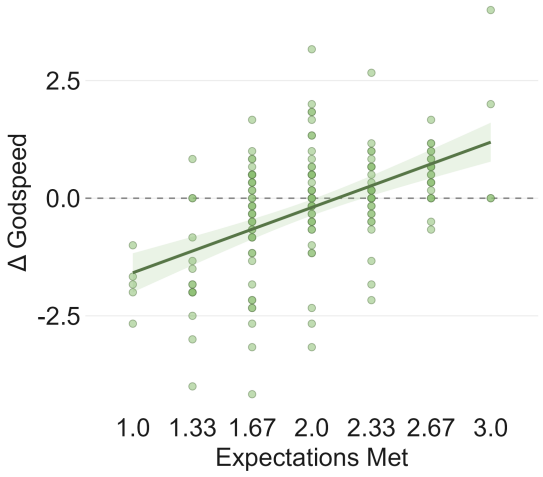


Figure 28: Godspeed change as a function of whether the model met task-level expectations (composite of three per-task ratings: fell short, met, exceeded). OLS fit with 95% CI band, N=162.